HIGH-DIMENSIONAL STATISTICS / CONVEX OPTIMIZATION

Sparse and Low-Rank Gaussian Graphical Models

A technical review of latent-variable model selection

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Scope and contribution

This report synthesizes the framework of Chandrasekaran, Parrilo, and Willsky [1]. My contribution is an exposition connecting the Schur complement, sparse–low-rank decomposition, geometric identifiability, and statistical recovery. The model, recovery results, and numerical studies reviewed here are attributed to that paper. This edition does not present a new estimator, an original consistency proof, or an independently executed experiment.

1 Why latent variables change the graph

A Gaussian graphical model describes conditional independence among jointly Gaussian variables. For covariance $\Sigma \succ 0$ and precision $K = \Sigma^{-1}$, an off-diagonal zero $K_{ij} = 0$ means that variables i and j are independent conditional on the remaining variables. Precision entries are not themselves correlation coefficients; the corresponding partial correlation is $-K_{ij}/\sqrt{K_{ii}K_{jj}}$.

In high dimensions, the sample covariance can be singular or unstable. Sparse precision estimation reduces complexity, but a sparse-only model can become misleading when unobserved factors influence many observed variables. In stock returns, for example, common market effects can induce widespread dependence.

Partition the joint precision matrix into observed (O) and hidden (H) blocks:

$$K^*_{(OH)} = \begin{pmatrix} K^*_O & K^*_{O,H} \\ K^*_{H,O} & K^*_H \end{pmatrix}.$$

The Schur complement gives the *marginal* precision of the observed vector:

$$\widetilde{K}^*_O = (\Sigma^*_O)^{-1} = \underbrace{K^*_O}_{S^*} - \underbrace{K^*_{O,H}(K^*_H)^{-1}K^*_{H,O}}_{L^*}. \tag{1}$$

Here S^* is the precision of X_O *conditional on* X_H. The matrix $L^* \succeq 0$ has rank at most the number of hidden variables and can be dense. Thus, the observed precision $S^* - L^*$ need not be sparse even when the conditional graph is sparse. Marginalization and conditioning are different operations; this distinction is central to the interpretation of the model.

2 A joint estimation objective

Let $\widehat{\Sigma}_O$ be the sample covariance of the observed variables. The regularized likelihood formulation reviewed in [1] is

$$\begin{aligned}(\widehat{S}, \widehat{L}) = \arg\min_{S,L} \; & -\log\det(S-L) + \operatorname{tr}\left[\widehat{\Sigma}_O(S-L)\right] \\ & + \lambda_n(\gamma\|S\|_1 + \operatorname{tr}(L)), \\ \text{subject to } & S - L \succ 0, \qquad L \succeq 0.\end{aligned} \tag{2}$$

The likelihood term fits the observed covariance. The entrywise ℓ_1 penalty encourages sparsity in S, while the trace penalty encourages low rank in L. For positive-semidefinite L, its trace equals its nuclear norm. The parameter λ_n controls overall regularization and γ controls the relative penalty on the sparse component.

The unconstrained regularizer $f_\gamma(S, L) = \gamma\|S\|_1 + \|L\|_*$ has dual norm

$$g_\gamma(S, L) = \max\left\{\frac{\|S\|_\infty}{\gamma}, \|L\|_2\right\}, \tag{3}$$

where $\|\cdot\|_\infty$ denotes the largest absolute matrix entry and $\|\cdot\|_2$ denotes the spectral norm. The placement of γ matters: the sparse-component norm is divided by γ.

3 When can the two components be separated?

Sparsity and low rank alone do not guarantee a unique decomposition. A matrix concentrated on a small set of entries can be both sparse and low rank. The geometric analysis asks whether local perturbations of the two structures can mimic one another.

For a sparse matrix S^* and a rank-r matrix $L^* = UDV^\top$, the relevant tangent spaces are

$$\Omega(S^*) = \{N : \operatorname{supp}(N) \subseteq \operatorname{supp}(S^*)\}, \tag{4}$$

$$T(L^*) = \{UY_1^\top + Y_2V^\top : Y_1, Y_2 \in \mathbb{R}^{p\times r}\}. \tag{5}$$

The latter expression uses the ambient matrix space; symmetric perturbations are used for the symmetric model. Local separability requires $\Omega \cap T = \{0\}$, rather than orthogonality of the two spaces.

The quantities

$$\mu(\Omega) = \max_{N\in\Omega,\, \|N\|_\infty\le 1} \|N\|_2, \qquad \xi(T) = \max_{N\in T,\, \|N\|_2\le 1} \|N\|_\infty$$

measure how strongly perturbations can concentrate under the two norms. One sufficient geometric criterion uses

$$\chi(\Omega, T, \gamma) = \max\left\{\frac{\xi(T)}{\gamma}, 2\mu(\Omega)\gamma\right\}. \tag{6}$$

If $\mu(\Omega)\xi(T) < 1/2$, choosing $\xi(T) < \gamma < 1/[2\mu(\Omega)]$ makes $\chi < 1$. This tangent-space condition is a useful starting point, but is not by itself the full set of assumptions needed for statistical consistency.

4 From geometric separation to statistical recovery

4.1 Likelihood curvature and stability

The likelihood adds a second layer of geometry. At the true marginal precision, the Fisher information acts on a perturbation Δ as

$$\mathcal{I}^*(\Delta) = \Sigma_O^* \Delta \Sigma_O^*.$$

The recovery analysis must control both the information retained within the structured tangent spaces and its leakage into complementary directions. It also controls how the tangent space of the low-rank variety changes near L^*. A very small nonzero singular value makes rank recovery less stable. These are stronger requirements than merely separating two spaces at one point.

4.2 A schematic reading of Theorem 4.1

Write $\xi = \xi(T)$ and $\mu = \mu(\Omega)$. Let σ be the smallest nonzero singular value of L^* and θ the smallest magnitude of a nonzero entry of S^*. Under the identifiability and Fisher-information assumptions of Proposition 3.3, with γ in its admissible range, Theorem 4.1 of [1] uses the following scaling conditions:

$$n \gtrsim \frac{p}{\xi^4}, \qquad \lambda_n \asymp \frac{1}{\xi}\sqrt{\frac{p}{n}}, \tag{7}$$

$$\sigma \gtrsim \frac{\lambda_n}{\xi^2}, \qquad \theta \gtrsim \frac{\lambda_n}{\mu}. \tag{8}$$

This is a schematic summary, not a replacement for the full theorem. The symbols $\gtrsim$ and $\asymp$ suppress factors involving the covariance scale and the Fisher-information constants. Those factors, as well as the admissible range of γ, must be retained when checking a particular model.

The theorem gives, with probability greater than $1 - 2e^{-p}$,

$$\operatorname{sign}(\widehat{S}) = \operatorname{sign}(S^*), \qquad \operatorname{rank}(\widehat{L}) = \operatorname{rank}(L^*), \tag{9}$$

$$g_\gamma(\widehat{S} - S^*, \widehat{L} - L^*) \lesssim \frac{1}{\xi}\sqrt{\frac{p}{n}}. \tag{10}$$

The first line describes *algebraic correctness*: recovering the signed sparse structure and the rank of the latent component. The second controls estimation error. Recovering a low-rank matrix does not identify a unique set of named latent factors or a unique factorization.

4.3 What the guarantee means in practice

- **Structure and estimation are distinct targets.** A small matrix error does not automatically imply exact edge or rank recovery when signals are weak.
- **Sample requirements depend on geometry.** The dimension p, incoherence quantity ξ, and sparse structure all matter; a sample-size condition depending only on p omits essential assumptions.
- **Theory guides rather than replaces validation.** Stability across regularization choices and performance on held-out observations are separate empirical questions.

5 Numerical evidence reported in the source paper

Provenance. The studies and numerical values in this section are from Section 6 of [1]. They are reviewed here, not reproduced by a new run.

5.1 Synthetic graphical models

The source paper studies three 36-node settings: two cycle-based models with two and three hidden variables, and a 6×6 grid with one hidden variable. Each setting uses 50 trials. The reported recovery curves illustrate how algebraic correctness improves as the sample-to-dimension ratio increases. Such finite simulations illustrate the theory; they do not prove consistency.

5.2 Stock-return example

The empirical example uses 216 monthly observations from 1990–2007 for 84 S&P 100 stocks. The paper compares a sparse-only fit with a latent-variable fit. Its reported comparison is summarized below.

Reported quantity	Sparse-only	Sparse + latent
Sparse edges	646	135
Latent components	0	5
Parameter count (paper's convention)	730	639
KL divergence relative to sample-covariance Gaussian	44.4	17.7

Source: Chandrasekaran et al. (2012), Section 6.2 and Figure 2.

The parameter count $84 + 135 + 5 \times 84 = 639$ follows the paper's convention. Lower divergence indicates better fit to its covariance-based reference, not demonstrated out-of-sample forecasting or trading performance.

For orientation, if the reference Gaussian is $\mathcal{N}(0, \widehat{\Sigma})$ and the fitted Gaussian is $\mathcal{N}(0, K^{-1})$, then

$$D_{\mathrm{KL}}\left(\mathcal{N}(0, \widehat{\Sigma}) \,\|\, \mathcal{N}(0, K^{-1})\right) = \tfrac{1}{2}\left[\operatorname{tr}(K\widehat{\Sigma}) - \log\det(K\widehat{\Sigma}) - p\right].$$

This formula states the direction explicitly; KL divergence is asymmetric.

6 Discussion and next steps

The decomposition distinguishes local conditional relationships from shared latent effects. Its use requires attention to Gaussian assumptions, structural separation, and regularization stability. Temporal dependence and nonstationarity also deserve investigation; graph edges alone do not establish causality.

A future implementation could compare support recovery, latent rank, estimation error, and held-out likelihood against a sparse-only baseline. This remains proposed follow-up work.

Figure note. Original image attachments were unavailable; see Figures 1–2 in the cited paper.

References

[1] V. Chandrasekaran, P. A. Parrilo, and A. S. Willsky (2012). Latent variable graphical model selection via convex optimization. *The Annals of Statistics*, 40(4), 1935–1967. doi:10.1214/11-AOS949. Open-access text.